```
"Extract the main words or phrases that best summarize the"
  "following sentence. Return only the words from the sentence in"
  "a space-separated format, without extra explanations or examples.\n\n"
  "Example 1:
  Sentence: \"The teacher explains a complex math problem.\"
  Keywords: teacher, explains, math problem\n"
  "Example 2:
  Sentence: \"A boy is playing soccer in the park.\"
  Keywords: boy, playing, soccer, park\n\n"
  "Now, extract the keywords from the following sentence:\n\"{}\""
```

Figure 3: The prompt template used for LLM-based keyword extraction in EVQAScore. The LLM is instructed to extract salient entities, actions, and attributes from the concatenated question–answer text.

A Additional Downstream Filtering Results

We provide additional downstream evaluation results to further validate the effectiveness of EVQAScore-based data filtering across a broader range of benchmarks and evaluation dimensions.

Results on Standard VideoQA Benchmarks. Table 4 reports the performance of Video-LLaVA, VideoLLaMA 2, and Qwen2-VL-7B on four standard VideoQA benchmarks (ActivityNet, MSRVTT, MSVD, and TGIF) using training subsets selected by different metrics. Across all three model architectures, EVQAScore-selected subsets consistently achieve superior accuracy and score compared to PAC-S++ under the same data budget. Notably, models trained on only 12.5% of the data selected by EVQAScore frequently outperform the full-data baseline, confirming that EVQAScore identifies higher-quality training samples that lead to more effective VideoLLM fine-tuning.

Results on VCGBench and MVBench. Table 5 further reports results on VCGBench and MVBench, which cover fine-grained video understanding dimensions (Correctness of Information, Detail Orientation, Contextual Understanding, Temporal Understanding, and Consistency) as well as multi-choice video comprehension. EVQAScore-selected subsets again yield consistently stronger performance across most metrics and model architectures. These results complement the findings in the main text and further demonstrate that metric performance on VAQ-Bench translates to effective data selection across diverse downstream evaluation settings.

B Case Study

We present a qualitative case study to illustrate how EVQAScore distinguishes answer quality in practice. As shown in Figure 4, the question asks about the action depicted in the video. Answer 1 provides a factually accurate response, while Answer 2 introduces hallucinated content that is inconsistent with the visual evidence. Our LLM-based keyword extraction method successfully identifies the hallucinated elements in Answer 2, enabling EVQAScore to penalize it accordingly. As a result, EVQAScore is the only metric among those evaluated that correctly determines Answer 1 to be superior.

This case study provides qualitative evidence that EVQAScore captures both semantic correctness and visual grounding, making it a reliable metric for evaluating VideoQA data quality.

C Keyword Extraction Prompt

As described in Section 4.2, EVQAScore uses an LLM to extract salient keywords from the concatenated question–answer text. Figure 3 shows the prompt template used for this extraction step. The prompt instructs the LLM to identify key entities, actions, attributes, and quantities from the input text while discarding redundant syntactic structures. This design ensures that the resulting keyword set captures the core semantics relevant to visual grounding, which is critical for accurate alignment scoring.

Table 4: Downstream validation: VideoLLM performance on ActivityNet, MSVD, MSRVTT, TGIF using training data selected by metrics evaluated on VAQ-Bench. Higher VAQ-Bench agreement translates to better downstream data filtering, demonstrating the practical utility of the benchmark.

Method	Model	Data Proportion	ActivityNet		MSVD		MSRVTT		TGIF	
			Acc.	Score	Acc.	Score	Acc.	Score	Acc.	Score
Baseline	**VideoLLaMA 2**	100%	45.33	2.27	59.84	2.98	56.75	2.82	48.28	2.39
PAC-S++		25%	42.68	2.17	63.82	3.18	56.60	2.80	55.83	2.76
PAC-S++		12.5%	46.68	2.36	74.39	3.71	60.74	3.02	55.61	2.76
EVQAScore		25%	49.54	2.47	70.82	3.53	62.49	3.10	55.95	2.77
EVQAScore		12.5%	**54.65**	**2.71**	**77.87**	**3.88**	**63.95**	**3.17**	**58.84**	**2.92**
Baseline	**Video-LLaVA**	100%	49.49	2.45	65.82	3.27	57.55	2.85	53.47	2.65
PAC-S++		25%	52.16	2.64	72.96	3.64	63.57	3.16	55.46	2.75
PAC-S++		12.5%	54.28	2.73	74.63	3.72	62.66	3.11	55.70	2.76
EVQAScore		25%	54.98	2.74	76.41	3.81	**65.41**	**3.25**	56.09	2.78
EVQAScore		12.5%	**56.56**	**2.83**	**77.62**	**3.87**	64.21	3.19	**58.17**	**2.88**
Baseline	**Qwen2-VL-7B**	100%	28.01	1.37	58.46	2.91	37.77	1.87	41.48	2.05
PAC-S++		25%	31.93	1.56	48.26	2.41	40.79	2.03	42.08	2.09
PAC-S++		12.5%	29.03	1.42	41.73	2.08	34.54	1.71	36.85	1.83
EVQAScore		25%	**32.68**	**1.61**	63.03	**3.14**	**47.07**	**2.34**	48.17	2.38
EVQAScore		12.5%	30.70	1.51	**63.05**	**3.14**	43.96	2.18	**48.40**	**2.39**

Table 5: Downstream validation: VideoLLM performance on VCGBench and MVBench using training data selected by metrics evaluated on VAQ-Bench. Higher VAQ-Bench agreement translates to better downstream data filtering, demonstrating the practical utility of the benchmark.

Method	Model	Data Proportion	VCGBench						MVBench
			CI	DO	CU	TU	CO	Avg.	Avg.
Baseline	**VideoLLaMA 2**	100%	2.15	1.79	2.43	1.52	2.41	2.06	30.85
PAC-S++		25%	1.79	1.49	2.11	1.55	2.18	1.82	32.26
PAC-S++		12.5%	1.84	1.55	2.14	1.43	2.13	1.82	29.97
EVQAScore		25%	**2.75**	**2.55**	**3.07**	2.02	**2.96**	**2.67**	**32.39**
EVQAScore		12.5%	2.69	**2.55**	3.05	**2.07**	2.89	2.65	31.46
Baseline	**Video-LLaVA**	100%	2.43	2.17	2.75	1.83	2.60	2.36	38.99
PAC-S++		25%	1.89	1.69	2.23	1.68	2.31	1.96	37.88
PAC-S++		12.5%	1.87	1.66	2.21	1.60	2.35	1.94	36.87
EVQAScore		25%	**2.63**	**2.48**	**2.96**	**1.99**	2.70	**2.55**	**39.69**
EVQAScore		12.5%	2.53	2.40	2.89	1.94	**2.73**	2.50	38.66
Baseline	**Qwen2-VL-7B**	100%	1.71	1.62	1.97	1.08	2.04	1.68	30.09
PAC-S++		25%	1.46	1.20	1.73	0.77	1.83	1.40	**31.78**
PAC-S++		12.5%	1.24	1.03	1.53	0.58	1.76	1.23	30.91
EVQAScore		25%	2.43	2.23	2.67	**1.45**	2.58	2.27	30.38
EVQAScore		12.5%	**2.48**	**2.31**	**2.71**	1.43	**2.60**	**2.31**	30.43

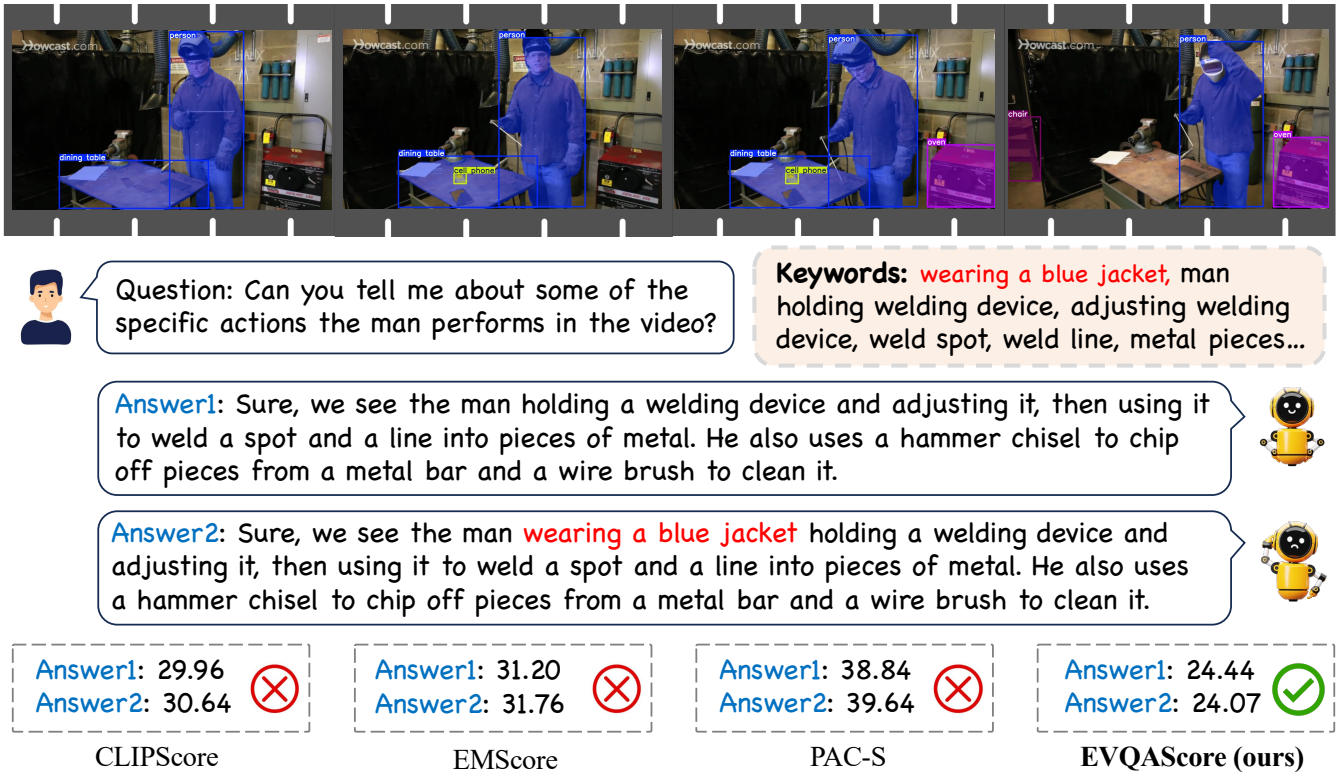

Figure 4: Answer1 is more accurate while Answer2 introduces hallucination. Our keyword extraction method successfully identifies hallucinations, making EVQAScore the only metric capable of determining that Answer1 is superior to Answer2.